

Lecture 1

Course Content

- **Introduction to Cloud Computing** (Definition, Evolution, Characteristics, Benefits)
- **Cloud Service Models** (SaaS, PaaS, IaaS, DaaS, FaaS)
- **Cloud Deployment Models** (Public, Private, Hybrid)
- **Cloud Computing Architecture** (Virtualization, Datacenters, Virtual Machines)
- **Virtualization Technology** (Type 1 and Type 2, VMs and Containers, Full & Para)
- **Cloud Storage** (Object, File & Block Storage), Providers, Backups, Archives
- **Cloud Security and Privacy** (Privacy and Compliance, IAM, ISO 27001, NIST)
- **Cloud Networking** (VPC, SDN, Load Balancing)
- **Cloud Application Development** (DevOps, CI/CD, Containerization & Kubernetes)
- **Current Trends in Cloud Computing**
- **Cloud Platforms & Tools** (AWS – EC2, S3, RDS, Lambda; Azure – ARM; GCP, etc.)

Slide 1: Introduction to Cloud Computing

- Cloud computing is the delivery of computing services over the internet
- Services include servers, storage, databases, networking, software
- Resources are provided on-demand
- Pay-as-you-go pricing model
- No need for physical infrastructure

Slide 2: Evolution of Cloud Computing

- Mainframe computing
- Client–server computing
- Distributed computing
- Grid computing

- Cloud computing
-

Slide 3: Characteristics of Cloud Computing

- On-demand self-service
 - Broad network access
 - Resource pooling
 - Rapid elasticity and scalability
 - Measured service
-

Slide 4: Benefits of Cloud Computing

- Cost efficiency
 - Scalability and flexibility
 - High availability
 - Disaster recovery
 - Reduced maintenance
-

Slide 5: Cloud Service Models

- Software as a Service (SaaS)
 - Platform as a Service (PaaS)
 - Infrastructure as a Service (IaaS)
 - Desktop as a Service (DaaS)
 - Function as a Service (FaaS)
-

Slide 6: SaaS, PaaS, IaaS

- **SaaS:** Ready-to-use applications
- **PaaS:** Platform for application development

- **IaaS:** Virtualized computing resources
 - Users access services via internet
 - Provider manages infrastructure
-

Slide 7: Cloud Deployment Models

- Public Cloud
 - Private Cloud
 - Hybrid Cloud
-

Slide 8: Public, Private, Hybrid Cloud

- **Public Cloud:** Shared resources, cost-effective
 - **Private Cloud:** Dedicated resources, high security
 - **Hybrid Cloud:** Combination of public and private
-

Slide 9: Cloud Computing Architecture

- Front-end (client side)
 - Back-end (cloud side)
 - Virtualization layer
 - Datacenters
 - Network connectivity
-

Slide 10: Virtualization Technology

- Virtualization enables multiple OS on one hardware
- Improves resource utilization
- Reduces hardware cost
- Enables isolation

Slide 11: Hypervisors

- Type 1 (Bare-metal hypervisor)
- Type 2 (Hosted hypervisor)
- Examples: VMware, Hyper-V, VirtualBox

Slide 12: VMs and Containers

- Virtual Machines run full OS
- Containers share host OS
- Containers are lightweight
- Faster deployment using containers

Slide 13: Types of Virtualization

- Full virtualization
- Paravirtualization
- Hardware-assisted virtualization

Slide 14: Cloud Storage

- Object storage
- File storage
- Block storage
- Scalable and durable

Slide 15: Cloud Storage Management

- Cloud storage providers
- Data backups

- Data recovery
 - Archival storage
-

Slide 16: Cloud Security and Privacy

- Data confidentiality
 - Data integrity
 - Data availability
 - Shared responsibility model
-

Slide 17: Security Standards and Compliance

- Identity and Access Management (IAM)
 - ISO 27001
 - NIST standards
 - Regulatory compliance
-

Slide 18: Cloud Networking

- Virtual Private Cloud (VPC)
 - Software Defined Networking (SDN)
 - Load balancing
 - Network security
-

Slide 19: Cloud Application Development

- DevOps practices
- Continuous Integration (CI)
- Continuous Deployment (CD)
- Automation tools

Slide 20: Containerization and Kubernetes

- Application packaging using containers
- Container orchestration
- Kubernetes for scaling and management
- High availability

Slide 21: Current Trends in Cloud Computing

- Serverless computing
- Edge computing
- Multi-cloud strategy
- AI and ML integration

Slide 22: Cloud Platforms & Tools

- **AWS:** EC2, S3, RDS, Lambda
- **Microsoft Azure:** ARM
- **Google Cloud Platform (GCP)**
- Popular cloud providers

Introduction to Cloud Computing

Cloud computing is a technology that allows users to access computing resources—such as **servers, storage, databases, software, and networking—over the Internet** instead of owning and maintaining physical hardware. These resources are provided by cloud service providers on a **pay-as-you-use** basis and can be accessed anytime, from anywhere.

Instead of installing software or storing data on a personal computer, users can use online services hosted in large **data centers**. **Cloud computing offers key features like on-demand access, scalability, flexibility, cost efficiency, and high availability.**

Example

A common example of cloud computing is **Google Drive**:

- Files are stored on Google's cloud servers, not on your local computer.
- You can access your documents from a laptop, mobile phone, or tablet.
- No need to buy extra storage hardware.
- Data is automatically backed up and synchronized.

Other examples include:

- **Gmail** (email service)
- **Amazon Web Services (AWS)** for hosting websites
- **Microsoft Azure** for application development
- **Netflix**, which streams content using cloud infrastructure

What is Cloud Computing?

Cloud computing means delivering **applications and IT infrastructure over the Internet** instead of running them on local computers or servers.

1. Cloud Computing: Apps and Infrastructure over the Internet

- Software, storage, servers, and networks are accessed online
- No need to install or maintain heavy hardware locally

Example: Gmail, Google Drive, Microsoft Office 365

2. Compute as a Service: Applications over the Internet

- Computing power and applications are provided as a service
- Users just use the application; infrastructure is managed by the provider

Example: Using AWS EC2 to run applications without owning servers

3. Utility Computing (Pay-As-You-Go)

- Similar to electricity or water billing
- You pay only for what you use (CPU, storage, bandwidth)

Example: Paying only for the hours a virtual server runs on AWS or Azure

Three New Aspects of Cloud Computing

1. Illusion of Infinite Computing Resources

- Resources appear unlimited
- Users can scale up or down anytime

Example: A website handling sudden traffic during sales

2. No Upfront Commitment

- No need to buy expensive hardware in advance
- Start small and expand when needed

Example: A startup launching an app without buying servers

3. Pay for Short-Term Usage

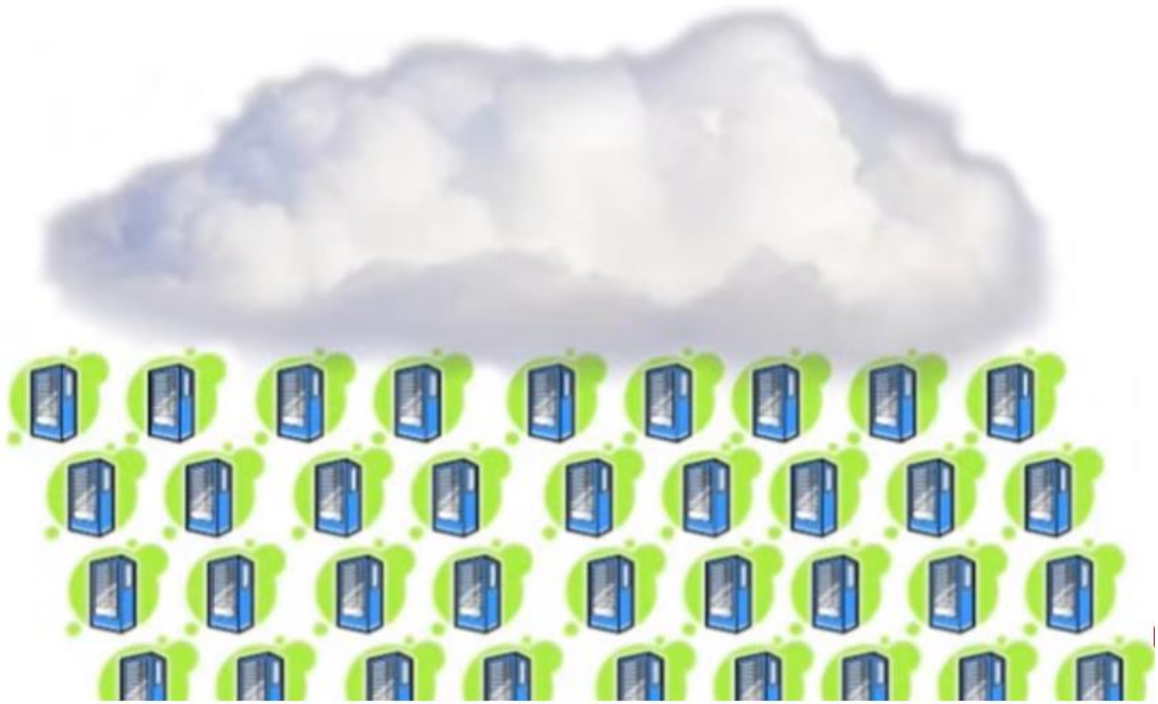
- Resources can be used temporarily
- Billed per hour, minute, or second

Example: Using cloud servers only during exams or peak business hours

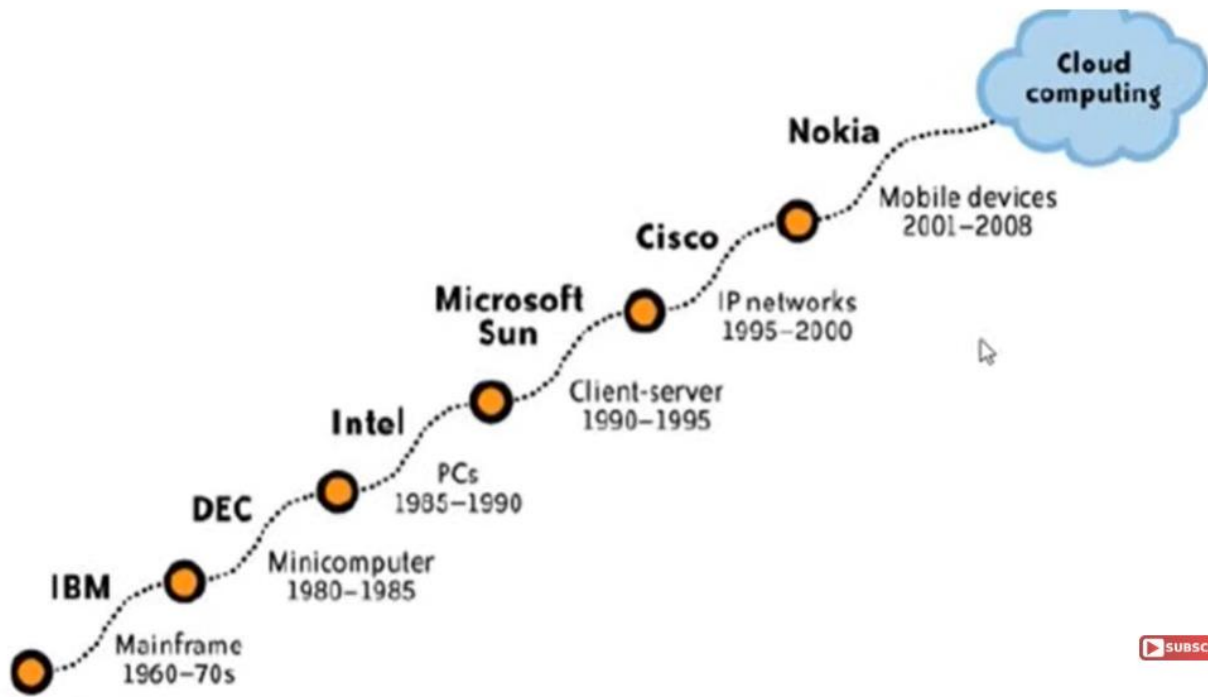
The Cloud is typically a large data-center



and very different from a PC...



Evolution of Computing (Explanation)



1. Mainframe Computers (1960s–1970s) – IBM

- Very large and expensive computers
- Used by governments and big organizations
- All processing was centralized
- Users accessed systems through terminals

2. Minicomputers (1980–1985) – DEC

- Smaller and cheaper than mainframes
- Used by universities and medium-sized companies
- Supported multiple users at the same time

3. Personal Computers (1985–1990) – Intel

- Computers became affordable for individuals
- Standalone systems used at homes and offices

- Each user had their own machine

4. Client–Server Computing (1990–1995) – Microsoft, Sun

- Servers handled data and processing
- Clients (PCs) requested services from servers
- Popular in enterprise networks

5. IP Networks / Internet Era (1995–2000) – Cisco

- Growth of the Internet and networking
- Data and services became globally accessible
- Enabled web-based applications

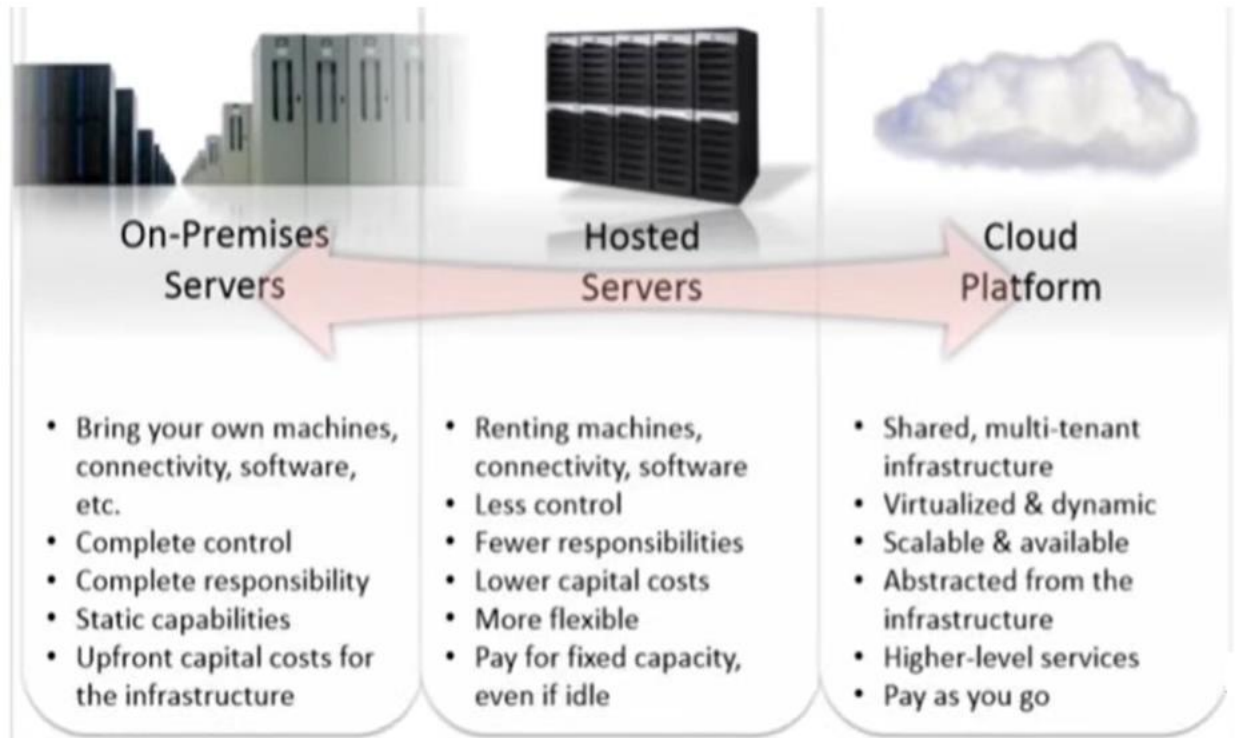
6. Mobile Computing (2001–2008) – Nokia

- Rise of mobile phones and portable devices
- Users accessed services anytime, anywhere
- Increased demand for online storage and apps

7. Cloud Computing (Present)

- Computing resources delivered over the Internet
- On-demand access to servers, storage, and applications
- Scalable, cost-effective, and flexible
- Examples: AWS, Azure, Google Cloud

IT infrastructure



1. On-Premises Servers

(Traditional setup – everything in your own office)

Explanation:

- The organization **buys and owns** all hardware and software.
- Servers are installed and managed **locally**.
- Full control, but also **full responsibility**.

Key Points:

- Bring your own machines, connectivity, and software
- Complete control and responsibility
- Fixed (static) capacity
- High upfront capital cost
- Maintenance, security, and upgrades are your job

Example:

A university maintains its own data center for student records and exam systems.

2. Hosted Servers

(Renting servers from a provider)

Explanation:

- Servers are hosted in a **third-party data center**.
- You rent the hardware instead of buying it.
- Some management is handled by the provider.

Key Points:

- Renting machines, software, and connectivity
- Less control than on-premises
- Fewer responsibilities
- Lower initial cost
- More flexible than on-premises
- Pay for fixed capacity, even when idle

Example:

A company rents dedicated servers from a hosting provider to run its website.

3. Cloud Platform

(Modern cloud computing model)

Explanation:

- Computing resources are provided **on-demand over the internet**.
- Infrastructure is shared and highly virtualized.
- You only pay for what you use.

Key Points:

- Shared, multi-tenant infrastructure
- Virtualized and dynamic resources
- Highly scalable and always available
- Infrastructure is abstracted (you don't manage hardware)
- Higher-level services (databases, analytics, AI)
- Pay-as-you-go pricing

Example:

Using **AWS EC2** for servers and **S3** for storage, scaling up during peak traffic and scaling down later.

Overall Comparison

Feature	On-Premises	Hosted	Cloud
Control	Very High	Medium	Low
Cost	High upfront	Medium	Pay-as-you-use
Scalability	Low	Limited	Very High
Responsibility	Full	Shared	Minimal